

📌 1. What is Collaborative Filtering?

Collaborative Filtering (CF) is a recommendation technique that makes predictions about a user's interests by **collecting preferences (ratings, likes, clicks) from many users**.

- The key assumption:

"Users who agreed in the past tend to agree in the future."

So, instead of analyzing the **content of items** (like text, genre, or features), CF looks at **user behavior (ratings, interactions)**.

Example:

- Suppose you and I both rated movies:
 - You: liked *Inception* ★★★★★ and *Interstellar* ★★★★★
 - Me: liked *Inception* ★★★★★ and *The Dark Knight* ★★★★★

Since our ratings for *Inception* are similar, the system assumes our tastes overlap. If I haven't rated *Interstellar*, CF may recommend it to me because **you liked it**.

📌 2. Types of Collaborative Filtering

1. **Memory-Based (Neighborhood methods):**

- Directly use the entire rating matrix (users × items).
- Find similar **users** or **items** using similarity measures.
- Example: User-based k-Nearest Neighbors (User-KNN).

2. **Model-Based:**

- Use ML models (like Matrix Factorization, SVD, Neural Nets) to learn latent features.
 - Scales better for large datasets.
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📌 3. How to Find Similar Users (Memory-Based Approach)

We compare user rating vectors with similarity metrics.

Step 1: Represent ratings as vectors

Suppose:

- User A ratings: [5, 4, ?, 2, ?]
- User B ratings: [4, 5, 3, ?, ?]

("?" = unrated)

Step 2: Compute similarity

Common choices:

- Cosine Similarity

$$\text{sim}(A, B) = \frac{\sum_i r_{A,i} \cdot r_{B,i}}{\sqrt{\sum_i r_{A,i}^2} \cdot \sqrt{\sum_i r_{B,i}^2}}$$

- Pearson Correlation (accounts for rating scale differences):

$$\text{sim}(A, B) = \frac{\sum_i (r_{A,i} - \bar{r}_A)(r_{B,i} - \bar{r}_B)}{\sqrt{\sum_i (r_{A,i} - \bar{r}_A)^2} \cdot \sqrt{\sum_i (r_{B,i} - \bar{r}_B)^2}}$$

Step 3: Select top-k similar users

Pick the **nearest neighbors** (e.g., top 5 most similar users).

Step 4: Predict ratings

For an item j not rated by user A:

$$\hat{r}_{A,j} = \bar{r}_A + \frac{\sum_{u \in N(A)} \text{sim}(A, u) \cdot (r_{u,j} - \bar{r}_u)}{\sum_{u \in N(A)} |\text{sim}(A, u)|}$$

